

UST'-UDA, Russian Federation

WMO: 305140

Lat: 54.167N

Lon: 103.017E

Elev: 427

StdP: 96.30

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -33.9	(d) -38.8	(e) 0.1	(f) -36.1	(g) -36.3	(h) 0.1	(i) -33.7	(j) 10.3	(k) -14.5	(l) 8.3	(m) -15.7	(n) 1.0	(o) 30	(p) 0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.7	(c) 28.5	(d) 18.4	(e) 26.4	(f) 17.4	(g) 24.4	(h) 16.7	(i) 19.7	(j) 25.5	(k) 18.7	(l) 24.1	(m) 17.7	(n) 22.5	(o) 3.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.5	22.1	16.7	12.6	21.0	15.9	11.9	20.3	58.7	25.3	55.2	24.2	52.0	22.7	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	7.8	6.7	-39.7	30.6	3.5	2.2	-42.2	32.2	-44.3	33.5	-46.2	34.8	-48.8	36.4
			-39.9	20.9	3.1	1.9	-42.2	22.2	-44.1	23.3	-45.8	24.3	-48.1	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-23.2	-20.5	-11.4	0.9	8.5	14.9	18.9	16.5	9.1	1.6	-8.9	-19.0		
	DBStd	15.36	6.84	6.37	6.27	3.96	4.04	3.91	3.27	3.04	3.65	4.17	6.41	6.66		
	HDD10.0	4691	1029	855	663	272	78	5	0	0	60	262	568	899		
	HDD18.3	7118	1288	1088	921	522	305	116	33	73	277	520	818	1157		
	CDD10.0	695	0	0	0	1	32	153	276	201	33	0	0	0		
	CDD18.3	80	0	0	0	0	0	13	51	16	0	0	0	0		
	CDH23.3	643	0	0	0	0	18	152	370	102	1	0	0	0		
	CDH26.7	146	0	0	0	0	2	38	93	13	0	0	0	0		
	WSAvg	2.7	1.9	1.7	2.4	3.2	3.3	2.9	2.7	2.5	3.1	3.2	3.1	2.0		
	PrecAvg	397	14	10	10	17	30	57	81	76	43	19	19	19		
(15) Precipitation	PrecMax	499	44	17	28	44	53	98	151	135	81	36	31	45		
	PrecMin	289	4	1	2	9	10	27	41	29	24	5	9	5		
	PrecStd	56	7	4	5	9	11	22	25	27	15	8	6	9		
	PrecStd	56	7	4	5	9	11	22	25	27	15	8	6	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.0	-0.9	5.1	16.2	25.7	30.7	31.1	28.6	21.5	13.6	4.1	-2.2		
		MCWB	-3.8	-2.1	2.1	8.2	13.5	17.6	19.6	18.7	15.4	8.0	1.2	-3.8		
	2%	DB	-8.0	-5.8	2.5	12.4	21.9	27.5	29.3	26.2	18.9	11.0	1.6	-5.3		
		MCWB	-9.0	-6.9	-0.2	5.7	11.9	16.5	19.1	17.6	13.2	6.8	-0.2	-6.2		
	5%	DB	-11.2	-8.7	0.6	9.7	19.2	24.8	27.2	24.1	16.6	9.1	-0.1	-7.3		
		MCWB	-11.9	-9.5	-1.4	4.7	10.8	15.4	18.3	17.1	11.9	5.7	-1.8	-8.0		
	10%	DB	-13.7	-11.2	-1.5	7.4	16.7	22.3	25.3	22.4	14.7	7.4	-1.4	-9.4		
		MCWB	-14.3	-11.9	-3.1	2.9	9.3	14.6	17.6	16.5	10.9	4.7	-2.7	-10.0		
	0.4%	WB	-3.7	-2.0	2.5	9.1	14.4	19.8	22.1	20.0	15.6	8.8	1.5	-3.7		
		MCDB	-2.9	-0.8	4.7	14.2	23.3	26.9	28.6	26.0	19.7	12.5	3.4	-1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-8.9	-6.8	0.1	6.7	12.7	17.9	20.5	18.9	14.0	7.2	0.0	-6.1		
		MCDB	-8.0	-5.9	2.1	11.5	19.8	24.4	26.5	24.3	17.5	10.3	1.7	-5.3		
	5%	WB	-11.9	-9.5	-1.4	4.8	11.2	16.7	19.4	18.1	12.5	6.2	-1.4	-7.9		
		MCDB	-11.3	-8.6	0.4	9.3	18.1	22.5	24.9	22.6	15.8	8.8	-0.3	-7.3		
	10%	WB	-14.2	-11.8	-3.0	3.3	9.7	15.5	18.5	17.1	11.3	4.9	-2.7	-9.9		
		MCDB	-13.7	-11.1	-1.6	6.9	16.2	20.8	23.5	21.3	14.2	7.2	-1.5	-9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.5	10.8	9.2	11.0	10.0	8.7	7.8	7.5	6.7	5.7	7.0		
		MCDBR	8.5	9.6	10.6	13.2	15.9	14.1	12.3	11.1	10.2	9.3	5.8	8.5		
		MCWBR	7.9	8.6	8.5	8.1	8.2	6.3	4.4	4.8	5.4	5.9	4.8	7.8		
		MCDBR	8.5	9.3	10.4	12.1	13.8	10.9	10.0	9.4	8.9	8.5	5.5	8.2		
	5% WB	MCWBR	8.0	8.4	8.4	7.9	7.7	5.8	4.7	4.9	5.4	5.8	4.8	7.6		
		MCWBR	8.0	8.4	8.4	7.9	7.7	5.8	4.7	4.9	5.4	5.8	4.8	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.312	0.340	0.383	0.430	0.433	0.433	0.394	0.347	0.330	0.305	0.283		
	taud		2.017	2.004	2.026	2.175	2.108	2.206	2.243	2.336	2.424	2.333	2.126	1.994		
	Ebn at Noon		625	745	818	839	815	817	809	823	822	741	603	530		
	Edh at Noon		73	105	130	129	146	134	127	110	88	76	66	61		
(44) All-Sky Solar Radiation	RadAvg		1.07	2.15	3.75	4.87	5.43	5.98	5.41	4.59	3.18	1.90	1.00	0.73		
	RadStd		0.06	0.10	0.15	0.35	0.46	0.38	0.49	0.27	0.28	0.16	0.09	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page